

The Internet of Things (IoT) is here today, and umpteen products are being announced to enable the enterprise, industrial space and home to be more efficient and automated. You are now able to talk to devices around your house including your lighting, appliances and even your doors! How? This interview explores what it really means to design for the IoT. Abhishek Mutha of *EFY* got in touch with industry experts from Broadcom, Intel, Infosys, IEEE, Bosch and ARM who shared their thoughts on various aspects related to the IoT like communication, security and its impact. Combining all their responses, we bring to you the bigger picture. Excerpts:

What about security? What security issues will be important for the IoT?

People are connecting devices that are important to them and have very sensitive data. This is a big change in the way people live connecting devices that have never been connected before.

If someone is of the notion that only the enterprise IP networks are prone to attacks, they are completely mistaken. Systems like smart grid attract more attention in terms of breaking into the system and creating havoc.

Before you were exposing yourself to the digital world through your smart phone or a computer, but the challenge now is that you will be exposing yourself through your house, which is intelligent and connected, to the digital world.

There are certain applications where security is not a big issue. Or you do not perceive it that way. For example, exchanging data for a game versus banking information. As we go forward, like we do with anything else in fiscal communication, we absolutely need to be very cognizant of security.

IPv6 offers better security and control capabilities compared to IPv4, but IPv6 security has not been field proven and testing has revealed vulnerabilities; therefore in short term, deploying applications using IPv6 represents a higher security risk.

The IoT will be about trust: generating trusted data from the device, getting information from it and its surroundings and providing services that you trust based on data. This takes the IoT from being about control to adding value.

Any examples showing the impact or the application of the IoT in industrial settings?

The industrial setting is one that is kind of underweighted in perspective of the IoT. We have actually seen Wi-Fi-connected manufacturing lines for several years now. There is a huge trend towards wearable devices. Wearables are an interesting concept. As things have gotten smaller, friendlier, specialised and simple, the wearable devices become more realistic and common.

The telecom towers are already using IoT concept in a very efficient way to manage diesel. They are managed by the IoT concept where sensors are involved that detect everything, be it the security aspect, level of diesel and communicate back to the system.

Around 80 per cent of fresh water we have in the world is used in farming, and we all know its inefficiency. A simple usage of sensors in the farms can help in reducing the amount of water being used. One area where there is lot of potential is smart grid. Today there are a lot of failures, losses and theft happening in power systems, so there is tremendous amount of potential for IoT technology to have an impact on the energy or power sector in general.

I think the way factories, industry and automation operate is going to get completely revolutionised. We are going to have very, very intelligent or smart factories. That means the operational costs are going to be more optimised. You are going to know more about what is going on in your production facilities early on.

How will the typical components manufacturing look like in the future? Components will only be manufactured in real time and in response to concrete orders. The components themselves will bear all information on technical requirements, customers and destination, allowing them to control the production process autonomously.

Let me give you an ARM-specific example – The Talking Book. The Talking Book from Literacy Bridge is a low-cost, handheld audio computer designed specifically for the learning needs of populations that cannot read in the poorest regions in the world. Recordings are stored on an internal microSD memory card whilst the built-in speaker enables group listening.

The complete interview can be read at <http://electronicsforu.com/newelectronics/interview/>